Quantitative Research Methods

Chapter 0: Precourse Refresher

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HSBI

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The course at a glance

	Lecture	Topic
▶	**0**	**Precourse — refresher of undergraduate statistics, algebra, Python**
	1	Ch. 1 + 2 — Introduction; what is statistical learning
	2	Ch. 2 — Accuracy, bias–variance, Bayes classifier, KNN
	3–4	Ch. 3 — Linear regression (estimation, inference, diagnostics)
	5–6	Ch. 4 — Classification (logistic, LDA/QDA, naive Bayes, ROC)
	7	Ch. 5 — Resampling (validation, CV, bootstrap)
	8	Ch. 6 — Model selection and regularisation (ridge, lasso)
	9	Ch. 7 — Moving beyond linearity (splines, GAMs)
	10	Ch. 8 — Tree-based methods (trees, forests, boosting)
	11	Ch. 10 — Deep learning (MLPs, CNNs, training)
	12	Ch. 13 — Multiple testing (FWER, FDR)

This refresher is *optional* and sits before Lecture 1. It assumes nothing beyond an introductory undergraduate course, and it only covers material the twelve lectures actually rely on. ▶ marks today's session.

Contents

Notation in this refresher

Symbol	Meaning
n, p	number of observations; number of variables
x_i, y_i	predictor and response value of observation i
$\bar{x}$, s, s^2	sample mean, sample standard deviation, sample variance
μ, σ, σ^2	population mean, standard deviation, variance
Q_1, Q_3, IQR	lower/upper quartile; interquartile range $Q_3 - Q_1$
r	sample correlation coefficient
$\mathrm{Cov}(X, Y)$	covariance of two random variables
$\mathrm{E}[X]$, $\mathrm{Var}(X)$	expected value and variance of X
$\hat{\theta}$	an estimate of the unknown quantity θ ("hat" = estimated)
$\mathrm{SE}(\hat{\theta})$	standard error: the SD of $\hat{\theta}$ across repeated samples
H_0, H_1	null and alternative hypothesis
α	significance level (tolerated false-positive rate)
X, y	$n \times p$ data matrix; n-vector of responses
$\boldsymbol{\beta}$, $\hat{\boldsymbol{\beta}}$	vector of true / estimated regression coefficients
∇f	gradient: the vector of partial derivatives of f

Why a refresher, and how to use it

The purpose

This course does not teach statistics from scratch — it teaches *statistical learning*. Everything here is assumed known from Lecture 1 onwards. One session, so that nobody is stuck on notation while we discuss the ideas that matter.

How to use it

- If a slide is obvious to you, that topic is fine — move on.
- If a slide is new, work its exercise before Lecture 1.
- Each section ends with *where this reappears in the course*, so you can see why it is worth the effort.

Learning objectives for this refresher

By the end of this session you will be able to:

- **Classify** variables by type and choose an appropriate summary and plot for each.
- **Compute and interpret** mean, median, variance, SD, quartiles, IQR, covariance and correlation.
- **Apply** the rules of probability, conditional probability and Bayes' theorem, and use expectation and variance rules.
- **Recognise** the normal, t, χ^2, F, Bernoulli, binomial and Poisson distributions and say where each is used.
- **Explain** sampling variability, standard errors, confidence intervals and p-values — and their standard misreadings.
- **Fit and interpret** a simple linear regression by hand.
- **Manipulate** vectors and matrices, and read the matrix form of the regression estimator.
- **Differentiate** a loss function and take a gradient-descent step.
- **Use** `numpy`, `pandas` and `matplotlib` for the basic tasks of the labs.

Do you actually need this session? A 10-question self-check

Answer honestly. Each question maps to one section of this deck.

1. Why is the median often reported for income, but the mean for test scores?
2. What does it mean that a variable has SD $= 0$?
3. Two variables have $r = 0$. Can they still be strongly related?
4. A test is 99% accurate and you test positive. Is there a 99% chance you are ill?
5. What is the difference between a standard deviation and a standard error?
6. What exactly is "95%" about a 95% confidence interval?
7. A p-value of 0.03: what is the probability that H_0 is true?
8. In $\hat{y} = 2 + 3x$, what does the 3 mean — in words, with units?
9. What are the dimensions of $\mathrm{X}^\top\mathrm{X}$ if X is $n \times p$?
10. What does the derivative of a loss function tell an optimisation algorithm to do?

Scoring

Comfortable with 8+? Skim this deck. Fewer than 5? Work through it with the exercises — it will pay for itself by Lecture 3.

Sources and references

Primary textbook (from Lecture 1 onwards)

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python.* Springer. Chapter 2 and the Chapter 3 appendix cover much of this refresher in condensed form.

If you want a fuller undergraduate treatment

Any introductory statistics text will do; the topics to look up are listed on the final "where to read more" slide. All figures in this deck were computed from the course data sets.

Outline

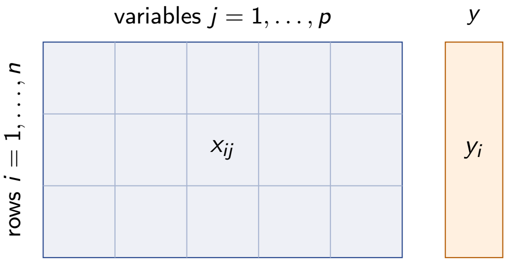
Roadmap of this refresher

How this session is organised

1. Data, variables and notation — the language.
2. Describing one variable, then two.
3. Probability, random variables and the standard distributions.
4. Sampling, estimation, confidence intervals, hypothesis tests.
5. Simple linear regression — the bridge into Chapter 3.
6. The linear algebra and calculus the later chapters use.
7. The Python toolkit of the labs.

Roughly one exercise every 20 minutes; solutions follow immediately.

The shape of a data set: n observations, p variables



X: the *data* (or design) matrix, $n \times p$

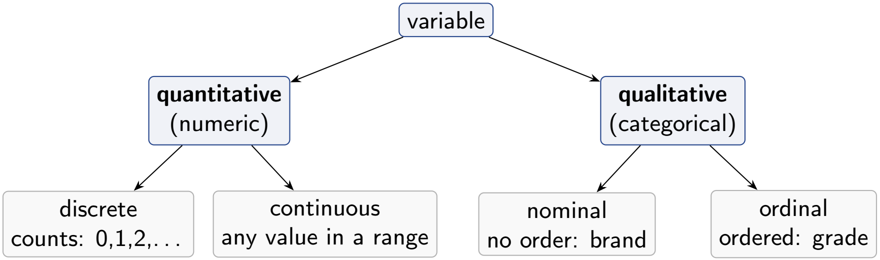
y: the response, one value per row

one *row* = one observation

Tidy data

One row per observational unit, one column per variable, one value per cell. Every method in this course expects data in this shape.

Variable types decide everything that follows



Type	Summarise with	Plot with
Quantitative	mean, SD, median, quartiles	histogram, boxplot, scatter
Qualitative	counts, proportions	bar chart, contingency table

Why the type matters in this course

The single most consequential distinction

The *response* type names the problem:

- quantitative $y \Rightarrow$ **regression** (Ch. 3, 6, 7),
- qualitative $y \Rightarrow$ **classification** (Ch. 4, 8, 10).

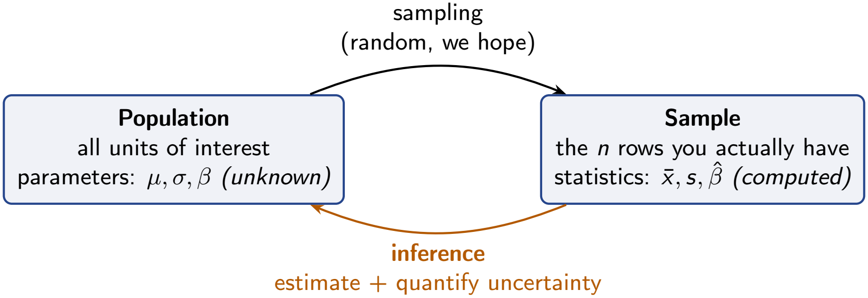
Qualitative *predictors* are not a problem

A categorical predictor enters as *dummy* (0/1) variables — one fewer than the number of categories, so “brand” with 3 levels costs 2 columns of X, not 1 (Chapter 3).

From the course data

`Wage`: `wage` is quantitative (regression, Ch. 7). `Default`: `default` is Yes/No (classification, Ch. 4). `Smarket`: `Direction` is Up/Down (classification, Ch. 4).

Population and sample: the loop this whole course lives in



Greek vs. Latin — and hats

Greek letters (μ, σ, β) denote unknown *population* quantities; Latin letters and hats ($\bar{x}$, s, $\hat{\beta}$) denote things *computed from data*. A hat always means "estimated".

Exercise 0.1 — Variable types and summaries [Concept]

Exercise

A university records, for each of n students: degree programme (Business, Engineering, Law); age in years; number of exam attempts; satisfaction on a 1–5 scale (1 = very poor, . . . , 5 = very good); and starting salary in euros.

1. Classify each of the five variables as quantitative (discrete / continuous) or qualitative (nominal / ordinal).
2. For each, name one appropriate numerical summary and one plot.
3. The university wants to predict starting salary. Is this a regression or a classification problem? What if it instead wants to predict whether a student graduates on time?
4. Someone computes the mean of "degree programme" by coding Business = 1, Engineering = 2, Law = 3, and reports 1.8. What is wrong with this?

Solution — Exercise 0.1

Solution

(1) and (2).

Variable	Type	Summary	Plot
Degree programme	qualitative, nominal	counts / proportions	bar chart
Age	quantitative, continuous	mean and SD (or median, IQR)	histogram, boxplot
Exam attempts	quantitative, discrete	mean, or median (skewed)	bar chart of counts
Satisfaction	qualitative, ordinal	median, quartiles	bar chart
Starting salary	quantitative, continuous	*median* (right-skewed)	histogram, boxplot

(3). Salary is quantitative $\Rightarrow$ **regression**. "Graduates on time" is Yes/No $\Rightarrow$ **classification**. The data are the same; the *response type* names the problem.

(4). The codes 1, 2, 3 are arbitrary labels, not quantities: Law is not "three times" Business, and the gaps are not equal or even meaningful. The mean 1.8 therefore has no interpretation — it would change if the categories were relabelled. Summarise nominal variables by counts and proportions. (In a model, such a variable becomes dummy variables, not a single numeric column.)

Outline

Three questions to ask of any single variable

Centre — spread — shape

1. **Centre**: where do typical values sit? (mean, median, mode)
2. **Spread**: how much do values differ? (SD, variance, range, IQR)
3. **Shape**: symmetric or skewed, one peak or several, outliers?

Never report a centre without a spread. "Average salary = €52,000" means something quite different with SD €3,000 than with SD €40,000.

Why the shape decides the summary

For symmetric data, mean and SD say it all. For skewed data (income, house prices, waiting times) the mean is dragged by the tail — report the median and IQR as well.

Measures of centre

Definitions

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \text{median} = \text{the middle value of the sorted data}, \qquad \text{mode} = \text{the most frequent v}$$

Measure	**Use when**	**Weakness**
Mean $\bar{x}$	data roughly symmetric; algebra needed	pulled by outliers
Median	data skewed or outlier-prone	ignores magnitudes; harder algebra
Mode	categorical data; finding peaks	may not be unique or exist

The mean is the balance point

$\sum_i (x_i - \bar{x}) = 0$ always: deviations above and below the mean cancel exactly. This identity is why squared deviations, not raw ones, are used to measure spread — and it reappears as the "residuals sum to zero" property in Chapter 3.

Measures of spread

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

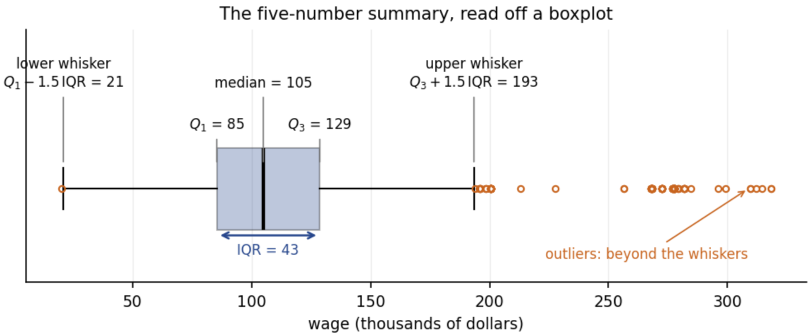
Reading it symbol by symbol

$(x_i - \bar{x})$ is how far observation i sits from the centre; squaring removes the sign and penalises large deviations; summing aggregates; dividing averages. The square root returns to the original units — which is why s, not s^2, is the number you report.

Why $n-1$ and not n?

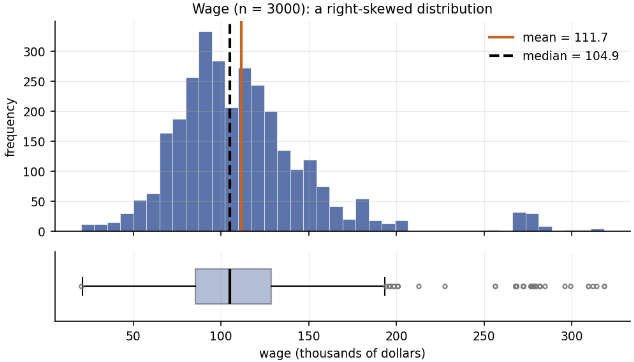
One degree of freedom was spent estimating $\bar{x}$ from the same data; dividing by $n-1$ makes s^2 unbiased for σ^2. The same bookkeeping returns in the t- and F-tests and in the effective-degrees-of-freedom arguments of Chapters 5–7.

Quantiles, the five-number summary, and the boxplot



The q-quantile is the value below which a fraction q of the data lies. $Q_1 = 25\%$, median $= 50\%$, $Q_3 = 75\%$; $\mathrm{IQR} = Q_3 - Q_1$ holds the middle half. The usual outlier rule flags points outside $[\,Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}\,]$ — a convention, not a law, and never a licence to delete data.

Shape: skewness read off a picture



`Wage` is right-skewed: a long upper tail pulls the mean (111.7) above the median (104.9). Rule of thumb: mean $>$ median suggests right skew, mean $<$ median left skew.

Standardisation: the *z*-score

Definition

$$z_i = \frac{x_i - \bar{x}}{s} \qquad \Longrightarrow \qquad \bar{z} = 0, \quad s_z = 1.$$

Why you will meet this constantly

A *z*-score says "how many standard deviations from the mean", so it is unit-free and comparable across variables. It matters in:

- **Ch. 6** — ridge and lasso penalise coefficients, so predictors *must* be standardised first, or the penalty depends on whether you measured in euros or thousands of euros;
- **Ch. 2, 12** — KNN and PCA use distances, which are dominated by whichever variable has the largest scale;
- **Ch. 10** — neural networks train far better on standardised inputs.

Worked example: a seven-point sample by hand

Seven observations computed by hand

The data: 2, 4, 4, 5, 7, 8, 12 (so $n = 7$).

Centre. $\sum x_i = 42$, so $\bar{x} = 42/7 = 6$. Sorted, the middle (4th) value is the median $= 5$. The mode is 4.

Spread. Deviations $x_i - \bar{x}$: $-4, -2, -2, -1, 1, 2, 6$; squares $16, 4, 4, 1, 1, 4, 36$, summing to 66. Hence

$$s^2 = \frac{66}{7-1} = 11, \qquad s = \sqrt{11} \approx 3.32.$$

Quartiles. Lower half $\{2, 4, 4\}$ gives $Q_1 = 4$; upper half $\{7, 8, 12\}$ gives $Q_3 = 8$; $\mathrm{IQR} = 4$. Fences: $4 - 6 = -2$ and $8 + 6 = 14$, so *no* outliers — 12 is large but not flagged.

Shape. $\bar{x} = 6 > 5 =$ median: mildly right-skewed, as the single large value 12 suggests.

Exercise 0.2 — Summaries and the effect of one outlier [Math]

Exercise

Seven service times (minutes) were recorded:

$$3, \quad 5, \quad 6, \quad 6, \quad 8, \quad 9, \quad 21.$$

1. Compute the mean, the median and the mode.
2. Compute s^2 and s. (Work with deviations from the mean.)
3. Compute Q_1, Q_3 and the IQR, and apply the $1.5 \times \mathrm{IQR}$ rule. Is 21 an outlier?
4. The 21 was a typo for 12. Recompute the mean and the median. Which moved more, and what general lesson does that illustrate?

Solution — Exercise 0.2

Solution

(1) Centre. $\sum x_i = 3 + 5 + 6 + 6 + 8 + 9 + 21 = 58$, so $\bar{x} = 58/7 \approx 8.29$. The 4th of seven sorted values is the median $= 6$. The mode is 6.
(2) Spread. Deviations from 8.29: $-5.29, -3.29, -2.29, -2.29, -0.29, 0.71, 12.71$; squares $27.9, 10.8, 5.2, 5.2, 0.1, 0.5, 161.6$, summing to 211.4. Hence $s^2 = 211.4/6 \approx 35.2$ and $s \approx 5.94$ minutes.
(3) Quartiles. Lower half $\{3, 5, 6\} \Rightarrow Q_1 = 5$; upper half $\{8, 9, 21\} \Rightarrow Q_3 = 9$; $\mathrm{IQR} = 4$. Upper fence $= 9 + 1.5(4) = 15 < 21$, so **yes**, 21 is flagged as an outlier. (Flagged means *investigate*, not *delete*.)
(4) After the correction. $\sum x_i = 49$, so $\bar{x} = 7$ — it fell by 1.29. The median is still 6: unchanged. The mean is *sensitive* to extreme values, the median is *resistant*. This is exactly why skewed variables (income, waiting times, house prices) are reported with the median, and why a single mistyped value can move a least-squares fit — squared errors magnify exactly this sensitivity (Chapter 3).

Outline

Covariance: do two variables move together?

Sample covariance

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Reading the sign

Each term is positive when x_i and y_i are on the *same* side of their means, negative otherwise. A positive sum means the variables tend to rise together; a negative sum means one rises as the other falls.

The problem with covariance

Its size depends on units: measuring x in cents rather than euros multiplies s_{xy} by 100, though nothing about the relationship changed. So we rescale it — and get the correlation.

Correlation: covariance made unit-free

Pearson correlation

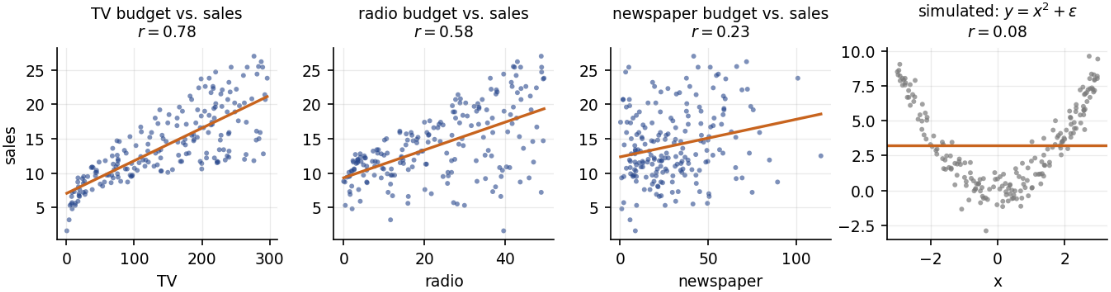
$$r = \frac{s_{xy}}{s_x s_y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}, \qquad -1 \le r \le 1.$$

Value	Meaning
$r = +1$ / -1	points lie exactly on an increasing / decreasing straight line
$r \approx 0$	no *linear* association (there may still be a strong curved one)
$\lvert r \rvert$ near 1	a tight linear pattern; near 0, a diffuse cloud

Three warnings

r measures *linear* association only; it is not resistant to outliers; and it says nothing about *causation* or about the *slope* of the relationship.

What r sees — and what it misses



The first three panels use `Advertising`: TV spend tracks sales closely ($r = 0.78$), newspaper spend barely ($r = 0.23$). The fourth is simulated data with an obvious, *strong* relationship $y = x^2 + \varepsilon$ — and $r = 0.08$. **Always plot the data.**

Correlation is not causation — the two classic traps

Confounding

Ice-cream sales and drowning deaths correlate strongly. Neither causes the other: *temperature* drives both. A variable that influences both X and Y is a **confounder**. Chapter 3 shows how adding it as a predictor can make a spurious effect disappear.

Simpson's paradox

An association can *reverse* once you condition on a group. A treatment can look worse overall yet be better in every subgroup, if the sicker patients were more likely to receive it. Aggregation can lie.

The habit to build

Before believing any two-variable claim, ask: *what third variable could produce this pattern?* This question is the entire reason multiple regression exists.

Two categorical variables: the contingency table

A study of 200 customers cross-tabulates a promotion against purchase:

	Bought	Did not buy	Total
Received promotion	45	55	100
No promotion	25	75	100
Total	70	130	200

- **Row proportions** are what you compare: 45% bought with the promotion vs. 25% without.
- If the variables were independent, the expected count in a cell is (row total $\times$ column total)/n: here $100 \times 70/200 = 35$ for the top-left cell. Observed 45 exceeds it.
- A χ^2 test formalises whether that gap is bigger than chance.

Where this returns

This table is a **confusion matrix** in disguise. In Chapter 4 the rows become "true class" and the columns "predicted class", and the same proportions become sensitivity and specificity.

Exercise 0.3 — Covariance, correlation and a slope [Math]

Exercise

Five stores report advertising spend x (thousands of euros) and sales y (thousands of units):

x	1	2	3	4	5
y	2	4	5	4	10

1. Compute $\bar{x}$ and $\bar{y}$.
2. Compute $S_{xx} = \sum(x_i - \bar{x})^2$, $S_{yy} = \sum(y_i - \bar{y})^2$ and $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$.
3. Compute r. Describe the association in one sentence.
4. The least-squares slope is $b_1 = S_{xy}/S_{xx}$. Compute it and state, in words and units, what it means.
5. Why do r and b_1 differ, even though both measure "how y moves with x"?

Solution — Exercise 0.3

Solution

(1) $\bar{x} = 15/5 = 3, \quad \bar{y} = 25/5 = 5.$
(2) Deviations: $x_i - \bar{x} = -2, -1, 0, 1, 2$ and $y_i - \bar{y} = -3, -1, 0, -1, 5.$

$$S_{xx} = 4 + 1 + 0 + 1 + 4 = 10, \quad S_{yy} = 9 + 1 + 0 + 1 + 25 = 36, \quad S_{xy} = 6 + 1 + 0 - 1 + 10 = 16.$$

(3) $r = \dfrac{16}{\sqrt{10 \cdot 36}} = \dfrac{16}{18.97} \approx 0.84$: a strong positive linear association — stores that spend more on advertising tend to sell more.
(4) $b_1 = 16/10 = 1.6$. *One extra thousand euros of advertising is associated with* 1.6 *thousand additional units sold*, on average, in this sample. Note "associated with", not "causes".
(5) They answer different questions. r is unit-free and bounded in $[-1, 1]$: it measures how *tightly* the points hug a line. b_1 carries units (units per thousand euros): it measures how *steeply* y changes with x. Formally $b_1 = r\, s_y/s_x$ — the same information, rescaled. A steep slope can be badly estimated (small $|r|$), and a tight relationship can have a nearly flat slope.

Outline

The rules you actually need

Basics

$0 \le P(A) \le 1; \quad P(\text{sure event}) = 1; \quad P(A^c) = 1 - P(A).$

Addition and multiplication

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \qquad P(A \cap B) = P(A \mid B)\, P(B).$$

Conditional probability and independence

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0); \qquad A \perp B \iff P(A \mid B) = P(A).$$

Conditioning is the whole game

Every classifier in this course estimates a conditional probability: $P(Y = \text{default} \mid X = \text{balance})$. "Learning" means turning data into a conditional probability you can act on.

Bayes' theorem: reversing the conditioning

The theorem

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}, \qquad P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

What it is for

You know $P(\text{evidence} \mid \text{cause})$ — how the test behaves for a sick person. You want $P(\text{cause} \mid \text{evidence})$ — whether *this* positive test means illness. Bayes' theorem converts one into the other, and the conversion depends on the **base rate** $P(A)$.

Where it reappears

Chapter 4 is Bayes' theorem applied four times over: the Bayes classifier, LDA, QDA and naive Bayes all compute $P(Y = k \mid X = x)$ from $P(X = x \mid Y = k)$ and the prior $\pi_k = P(Y = k)$.

Worked example: why a 95% accurate test can mislead

Screening for a rare condition

Prevalence $P(D) = 1\%$; sensitivity $P(+ \mid D) = 90\%$; specificity $P(- \mid D^c) = 95\%$, so $P(+ \mid D^c) = 5\%$. You test positive.

$$P(+) = \underbrace{0.90 \times 0.01}_{\text{true positives } = 0.009} + \underbrace{0.05 \times 0.99}_{\text{false positives } = 0.0495} = 0.0585.$$

$$P(D \mid +) = \frac{0.009}{0.0585} \approx 0.154.$$

Only about **15%**: the healthy group is 99 times larger, so its 5% false-positive rate produces five times more positives than the ill group does.

The lesson for Chapter 4

With rare events, high accuracy coexists with mostly-wrong positives — which is why Chapter 4 reports sensitivity, specificity and ROC curves rather than one accuracy number.

Random variables, expectation and variance

Definitions (discrete case)

$$\mathrm{E}[X] = \sum_{x} x\,P(X = x), \qquad \mathrm{Var}(X) = \mathrm{E}\big[(X - \mathrm{E}[X])^2\big] = \mathrm{E}[X^2] - (\mathrm{E}[X])^2.$$

Rules you will use constantly

$$\mathrm{E}[aX + b] = a\,\mathrm{E}[X] + b, \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X),$$
$$\mathrm{E}[X + Y] = \mathrm{E}[X] + \mathrm{E}[Y] \text{ (always)}, \qquad \mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \textit{ only if } X \perp Y.$$

Read the rules carefully

Expectation is linear *without conditions*. Variance is not: adding a constant does not change spread (b vanishes), scaling squares it (a^2), and variances add only under independence. The bias–variance decomposition in Chapter 2 is these rules applied to $\hat{f}(x)$.

Exercise 0.4 — Conditional probability and Bayes [Math]

Exercise

A bank's fraud filter flags transactions. Historically 0.5% of transactions are fraudulent. The filter flags 95% of fraudulent transactions, and also flags 2% of legitimate ones.

1. What proportion of *all* transactions get flagged?
2. A transaction is flagged. What is the probability that it is actually fraudulent?
3. Out of 100,000 transactions, fill in the four counts (fraud or not $\times$ flagged or not). Which cell dominates the flagged column?
4. The bank wants $P(\text{fraud} \mid \text{flag})$ above 50%. Holding the 95% detection rate fixed, what false-positive rate would be required? Comment on whether that is realistic.

Solution — Exercise 0.4

Solution

(1) With $P(F) = 0.005$:

$$P(\text{flag}) = 0.95(0.005) + 0.02(0.995) = 0.00475 + 0.0199 = 0.02465 \approx 2.5\%.$$

(2) $P(F \mid \text{flag}) = 0.00475/0.02465 \approx \mathbf{19\%}$: four of five flags are false alarms, despite a 95% detection rate.
(3) Per 100,000 transactions:

	Flagged	**Not flagged**	**Total**
Fraudulent	475	25	500
Legitimate	1,990	97,510	99,500
Total	2,465	97,535	100,000

The 1,990 false positives dominate: the legitimate group is 199 times larger, so even a small error rate on it swamps the true positives.
(4) Require $0.00475/(0.00475 + f \cdot 0.995) \geq 0.5$, i.e. $f \leq 0.0048$ — a false-positive rate below **0.48%**, four times stricter than today's 2%, and reachable only by flagging far less aggressively, which costs detection. That trade-off, drawn as a curve, is the ROC curve of Chapter 4.

Outline

The discrete trio

Bernoulli(π) — one yes/no trial

$P(X = 1) = \pi$, $P(X = 0) = 1 - \pi$; $\mathrm{E}[X] = \pi$, $\mathrm{Var}(X) = \pi(1 - \pi)$. *Where:* the response in every classification problem (Ch. 4, 8, 10).

Binomial(n, π) — number of successes in n independent trials

$P(X = k) = \binom{n}{k}\pi^k(1 - \pi)^{n-k}$; $\mathrm{E}[X] = n\pi$, $\mathrm{Var}(X) = n\pi(1 - \pi)$. *Where:* counting correct classifications; the bootstrap (Ch. 5).

Poisson(λ) — counts per unit of time or space

$P(X = k) = e^{-\lambda}\lambda^k/k!$; $\mathrm{E}[X] = \mathrm{Var}(X) = \lambda$. *Where:* Poisson regression on `Bikeshare` (Ch. 4).

Note the variance of a Bernoulli

$\pi(1 - \pi)$ is largest at $\pi = 0.5$ and vanishes at 0 or 1: a near-certain event is nearly noiseless. This is why classification is hardest near the decision boundary.

The normal distribution

$X \sim N(\mu, \sigma^2)$

Symmetric and bell-shaped, fully determined by μ and σ. Standardising gives $Z = (X - \mu)/\sigma \sim N(0, 1)$.

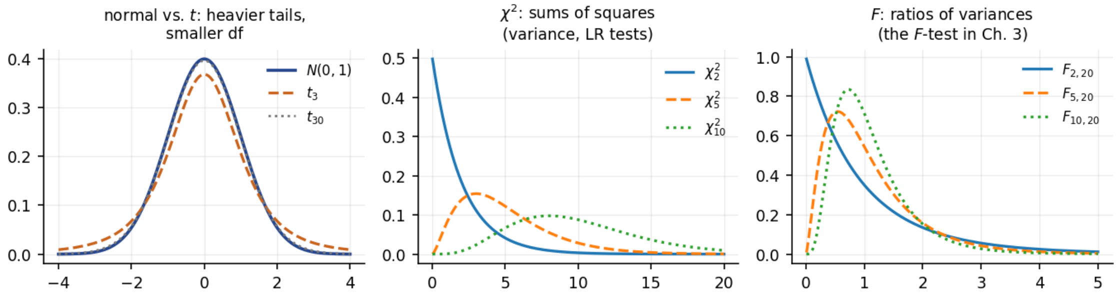
The 68–95–99.7 rule

About 68% of the mass lies within 1σ of μ, 95% within 2σ (more precisely 1.96σ), and 99.7% within 3σ. Memorising 1.96 saves you a table lookup for the rest of the course.

What is and is not assumed

Regression does *not* require X or y to be normal. Normality is assumed for the *errors* — and even then mainly to justify t- and F-tests in small samples. With large n the CLT does the work instead.

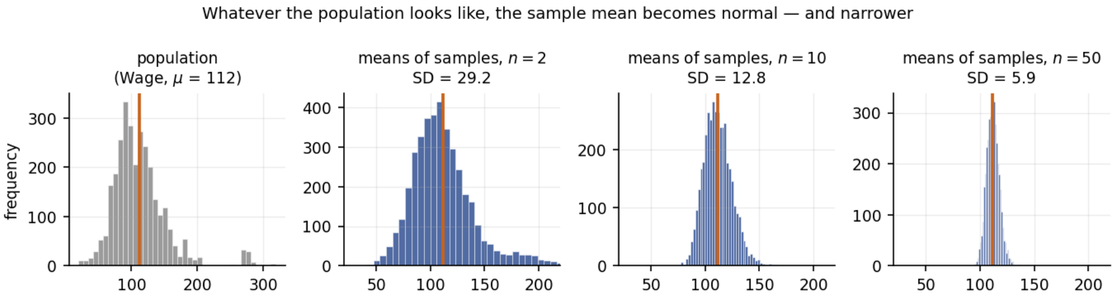
Three distributions that come from the normal



Distribution	Arises as	Used for
t_ν	normal mean with σ *estimated*	t-tests, CIs on coefficients (Ch. 3)
χ^2_ν	sum of ν squared standard normals	variance, likelihood-ratio tests
F_{d_1,d_2}	ratio of two scaled χ^2	the F-test for several coefficients (Ch. 3)

As the degrees of freedom grow, $t_\nu \to N(0,1)$: with $\nu > 30$ the difference is negligible, which is why large-sample rules of thumb use 1.96.

The central limit theorem



Statement

For a random sample of size n from *any* population with mean μ and finite variance σ^2, $\bar{X} \approx N(\mu, \sigma^2/n)$ for large n. The spread of $\bar{X}$ shrinks like $1/\sqrt{n}$ — quadrupling the sample halves the uncertainty.

Outline

Standard deviation vs. standard error

The distinction students most often blur

- The **standard deviation** s describes the spread of the *data*; more data does not shrink it.
- The **standard error** $\mathrm{SE}(\bar{x}) = s/\sqrt{n}$ describes the spread of an *estimate* over repeated samples; more data *does* shrink it.

Numbers

Wages with $s = 40$ and $n = 100$: $\mathrm{SE}(\bar{x}) = 40/10 = 4$. With $n = 1600$: $\mathrm{SE} = 40/40 = 1$. The people are just as variable as before; our estimate of their average is four times more precise.

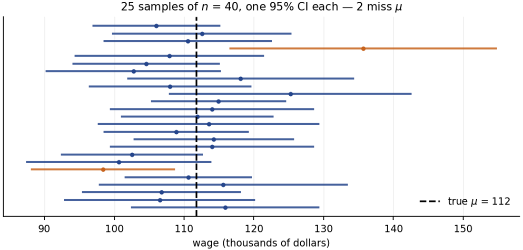
The habit

Report every estimate — a coefficient, an error rate, a CV score — *with* its standard error. A number without an uncertainty is not a result.

Confidence intervals

The usual interval for a mean

$$\bar{x} \pm t_{n-1,\,0.975} \cdot \frac{s}{\sqrt{n}} \qquad \left(\approx \bar{x} \pm 1.96\,\mathrm{SE} \text{ for large } n\right).$$



Each horizontal bar is one 95% CI from one sample. Two of the 25 miss the true μ — which is exactly what "95%" promises.

What a confidence interval does and does not say

The correct reading

"95%" is a property of the *procedure*: repeat the whole sampling-and-computing exercise many times, and about 95% of the intervals produced would contain the true parameter.

Three wrong readings

1. "95% probability that μ lies in *this* interval." — μ is fixed; this interval either contains it or not.
2. "95% of the data lie in the interval." — no: that is a *prediction* interval, which is far wider.
3. "It contains 0, so the effect is zero." — no: the data cannot rule zero out. Absence of evidence is not evidence of absence.

Width tells you as much as position

Wide interval around a large estimate: "possibly big — we cannot tell". Narrow interval around a small one: "genuinely small".

Outline

The logic of a test in five steps

1. State H_0 (the sceptical default: "no effect", $\beta_j = 0$) and H_1.
2. Choose α, the false-positive rate you will tolerate (conventionally 0.05).
3. Compute a **test statistic**: the estimate, measured in standard errors, e.g. $t = (\hat{\theta} - \theta_0)/\mathrm{SE}(\hat{\theta})$.
4. Compute the p-**value**: the probability, *if H_0 were true*, of a statistic at least this extreme.
5. Reject H_0 if $p < \alpha$; otherwise fail to reject. Then report the *estimate and its CI*, not just the verdict.

The test statistic is a signal-to-noise ratio

t asks: how large is the effect *relative to how precisely we measured it*? Every t-value in every regression output in this course is exactly this ratio.

Two ways to be wrong

	H_0 is true	H_0 is false
Reject H_0	Type I error (prob. α)	correct (power $= 1 - \beta$)
Do not reject	correct	Type II error (prob. β)

- Lowering α buys fewer false positives at the cost of more false negatives. There is no setting that avoids both.
- **Power** rises with the true effect size, with n, and with lower noise. Underpowered studies mostly produce misleading results.

The same table under different names

In Chapter 4 this becomes the confusion matrix (false positive / false negative); in Chapter 13 the columns are relabelled to define the family-wise error rate and the false discovery rate. It is one idea wearing three costumes.

What a p-value is not

Four standard misreadings

1. "*$p = 0.03$ means a 3% chance that H_0 is true.*" No: it is a probability computed *assuming* H_0 — it says nothing about $P(H_0 \mid \text{data})$.
2. "*$p > 0.05$ proves no effect.*" No: it may only mean the sample was too small to detect one.
3. "*$p = 0.001$ means a big effect.*" No: with huge n, trivial effects get tiny p-values. Significance $\neq$ importance.
4. "*I tested 20 things and one was significant at 5%.*" Expected by chance alone — see Chapter 13.

The professional habit

Report the estimate, its confidence interval, and the p-value — in that order of importance. Effect size first, uncertainty second, verdict last.

Worked example: a one-sample t-test

Has average handling time changed?

Historically a process took $\mu_0 = 100$ seconds. A sample of $n = 100$ recent cases gives $\bar{x} = 103$ and $s = 15$.

$$\mathrm{SE}(\bar{x}) = \frac{15}{\sqrt{100}} = 1.5, \qquad t = \frac{103 - 100}{1.5} = 2.00 \quad (\mathrm{df} = 99).$$

Two-sided $p \approx 0.048 < 0.05$: reject H_0 at the 5% level.
The more useful statement. The 95% CI is $103 \pm 1.98(1.5) = [100.03,\ 105.97]$: the increase is somewhere between 0.03 and 6 seconds — detectable, yet possibly far too small to matter operationally.

CI and test agree by construction

A two-sided test rejects at level α exactly when the $(1 - \alpha)$ CI excludes the null value. The CI simply says more.

Exercise 0.5 — Reading an inference [Concept]

Exercise

A colleague reports: "We estimated the effect of a training programme on productivity. The estimate is $+2.4$ units with standard error 1.5, giving $t = 1.6$ and $p = 0.11$. We conclude the programme has no effect."

1. Compute the approximate 95% confidence interval.
2. Is the conclusion "no effect" justified? Answer using the interval.
3. The team repeats the study with four times the sample size and obtains the same estimate $+2.4$. What is the new standard error (assuming the same variability), and the new t?
4. In a second study of 40 outcome variables, three came out significant at the 5% level. Roughly how many would you expect by chance if none of the 40 had any effect?

Solution — Exercise 0.5

Solution

(1) $2.4 \pm 1.96(1.5) = 2.4 \pm 2.94 = [-0.54,\ 5.34]$.
(2) No. The interval does contain 0, so a zero effect cannot be ruled out — but it also contains $+5.3$, a substantial benefit. The correct statement is "this study cannot distinguish between no effect and a sizeable positive effect": the data are *inconclusive*, not evidence of absence. Failing to reject H_0 is never the same as accepting it.
(3) The standard error scales as $1/\sqrt{n}$, so four times the data halves it: $\mathrm{SE} = 0.75$, and $t = 2.4/0.75 = 3.2$ ($p \approx 0.001$). The new CI is $2.4 \pm 1.47 = [0.93,\ 3.87]$ — same estimate, now clearly non-zero. *Significance is as much about sample size as about the effect.*
(4) With 40 true nulls and $\alpha = 0.05$, we expect $40 \times 0.05 = \mathbf{2}$ false positives, and $P(\text{at least one}) = 1 - 0.95^{40} \approx 0.87$. Three "discoveries" is therefore unremarkable. Correcting for this is the entire content of Chapter 13 (Bonferroni, Holm, Benjamini–Hochberg).

Extended Exercise 0.1 — From data to a defensible claim [Math]

Extended exercise (15 min)

A retailer measures the basket value (euros) of $n = 64$ customers after a store redesign. The sample gives $\bar{x} = 48.5$ and $s = 12.0$. The pre-redesign average was $\mu_0 = 45.0$.

1. Compute $\mathrm{SE}(\bar{x})$ and a 95% confidence interval for the true post-redesign mean (use 1.96).
2. Test $H_0 : \mu = 45$ against $H_1 : \mu \neq 45$ at $\alpha = 0.05$. Report t and your decision, and state how the CI would have told you the same thing.
3. Management says an increase is worth acting on only if it exceeds €5. Does the evidence support acting? Answer using the interval, not the p-value.
4. How large would n have to be for the CI half-width to shrink to €1.00 (same s)?
5. The redesign coincided with a marketing campaign. What does that do to the causal interpretation, and what design would fix it?

Solution — Extended Exercise 0.1 (1/2)

Solution

(1) Standard error and interval.

$$\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5.$$

$$95\%\ \mathrm{CI} = 48.5 \pm 1.96(1.5) = 48.5 \pm 2.94 = [\,45.56,\ 51.44\,].$$

(2) The test.

$$t = \frac{\bar{x} - \mu_0}{\mathrm{SE}} = \frac{48.5 - 45.0}{1.5} = \frac{3.5}{1.5} = 2.33 \quad (\mathrm{df} = 63).$$

Two-sided $p \approx 0.023 < 0.05$, so we reject H_0: the mean basket value differs from €45. The CI says the same thing without a separate calculation — it excludes 45. This equivalence is exact for a two-sided test at the matching level.

Solution — Extended Exercise 0.1 (2/2)

Solution

(3) The decision that management actually asked about. The relevant question is not "is the increase non-zero" but "is it above €5". The increase is estimated at $48.5 - 45.0 =$ €3.50, with CI $[45.56, 51.44] - 45 = [\,0.56,\ 6.44\,]$. The interval straddles 5: the data are consistent with an increase of €1 and with one of €6. So the evidence does *not* support acting — and note that the p-value alone (0.023, "significant") would have suggested the opposite. *Statistical significance answered the wrong question.*

(4) Required sample size. Half-width $1.96 \cdot 12/\sqrt{n} \leq 1.00$ requires $\sqrt{n} \geq 1.96 \cdot 12 = 23.52$, so $n \geq 553.2$, i.e. $n =$ **554**. Nearly nine times the data for a threefold gain in precision — the $1/\sqrt{n}$ law is unforgiving, and it is the same law that makes learning curves flatten in Chapter 2.

(5) Causality. The campaign is a **confounder**: the redesign and the campaign changed together, so their effects cannot be separated from these data — they are *confounded* in the technical sense. Fixes, in descending order of quality: randomise stores to redesign / no redesign while running the campaign everywhere; or use control stores with the campaign but no redesign (difference-in-differences); or, weakest, adjust for campaign exposure in a regression — which only works for confounders you thought to measure.

Outline

The model and the criterion

The model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \mathrm{E}[\varepsilon_i] = 0.$$

Least squares

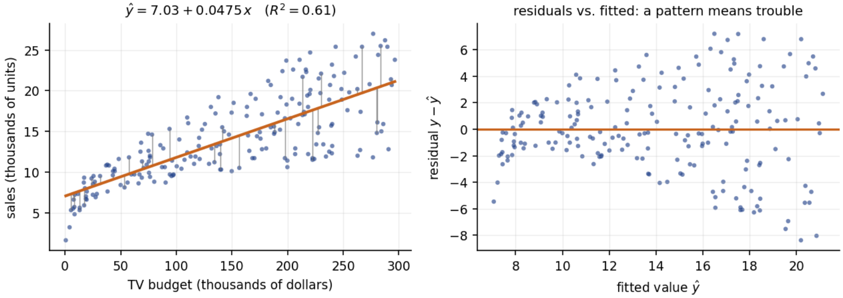
Choose b_0, b_1 minimising the residual sum of squares

$$\mathrm{RSS}(b_0, b_1) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2, \qquad b_1 = \frac{S_{xy}}{S_{xx}}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

Two things to notice now

(i) The fitted line always passes through $(\bar{x}, \bar{y})$. (ii) Squared errors are chosen for tractability, not sanctity — they make the algebra closed-form, and they make outliers expensive.

Fitted values, residuals, and R^2



Left: the fit on `Advertising`; the grey segments are the residuals $e_i = y_i - \hat{y}_i$ that least squares squares and adds up. Right: a residual plot — structure here (curvature, a funnel) means the model is missing something. Both plots return in Chapter 3.

Decomposing the variation

The three sums of squares

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{explained}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{RSS}}, \qquad R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

How to read R^2

The fraction of the variation in y that the model accounts for. $R^2 = 0.61$ means "61% of the variability in sales is accounted for by TV spend". In simple regression $R^2 = r^2$ exactly.

Do not over-read it

A high R^2 does not make a model correct, useful or causal, and a low one does not make it worthless. Adding any predictor never decreases R^2 — which is why Chapter 6 needs adjusted R^2, C_p, AIC, BIC and cross-validation.

Extended Exercise 0.2 — Regression by hand, end to end [Math]

Extended exercise (15 min)

Four branches report weekly advertising spend x (€000s) and sales y (€000s):

x	2	4	6	8
y	5	8	12	15

1. Compute $\bar{x}$, $\bar{y}$, S_{xx}, S_{xy} and S_{yy}.
2. Compute b_1 and b_0, and write out the fitted line.
3. Compute the four fitted values and residuals. Verify that the residuals sum to (approximately) zero, and explain why they must.
4. Compute RSS, TSS and R^2. Confirm $R^2 = r^2$.
5. Interpret b_0 and b_1 in words *with units*. Is the interpretation of b_0 trustworthy here?
6. Predict sales at $x = 5$ and at $x = 40$. Which prediction would you defend, and why?

Solution — Extended Exercise 0.2 (1/2)

Solution

(1) Sums. $\bar{x} = 20/4 = 5$, $\bar{y} = 40/4 = 10$. Deviations $x_i - \bar{x} = -3, -1, 1, 3$ and $y_i - \bar{y} = -5, -2, 2, 5$.

$$S_{xx} = 9 + 1 + 1 + 9 = 20, \quad S_{xy} = 15 + 2 + 2 + 15 = 34, \quad S_{yy} = 25 + 4 + 4 + 25 = 58.$$

(2) Coefficients.

$$b_1 = \frac{34}{20} = 1.7, \qquad b_0 = 10 - 1.7(5) = 1.5, \qquad \hat{y} = 1.5 + 1.7x.$$

(3) Fitted values and residuals.

x_i	2	4	6	8
$\hat{y}_i = 1.5 + 1.7x_i$	4.9	8.3	11.7	15.1
$e_i = y_i - \hat{y}_i$	$+0.1$	-0.3	$+0.3$	-0.1

The residuals sum to 0. This is forced by the first normal equation: setting $\partial \mathrm{RSS}/\partial b_0 = 0$ gives $\sum_i e_i = 0$ exactly. It is a property of the *fitting method*, not evidence that the model is right.

Solution — Extended Exercise 0.2 (2/2)

Solution

(4) Fit quality. $\mathrm{RSS} = 0.1^2 + 0.3^2 + 0.3^2 + 0.1^2 = 0.01 + 0.09 + 0.09 + 0.01 = 0.20$, and $\mathrm{TSS} = S_{yy} = 58$. So

$$R^2 = 1 - \frac{0.20}{58} = 0.99655.$$

Check: $r = S_{xy}/\sqrt{S_{xx}S_{yy}} = 34/\sqrt{20 \cdot 58} = 34/34.06 = 0.99827$, and $r^2 = 0.99655$. ✓

(5) Interpretation. $b_1 = 1.7$: each additional €1,000 of weekly advertising is associated with €1,700 more weekly sales, on average. $b_0 = 1.5$: predicted sales of €1,500 at *zero* advertising. Treat b_0 with care — $x = 0$ is outside the observed range $[2, 8]$, so it is an extrapolation, and here it is merely the intercept that makes the line fit, not a meaningful "baseline".

(6) Predictions. At $x = 5$: $\hat{y} = 1.5 + 8.5 = 10.0$ — defensible, since 5 sits inside the observed range (interpolation). At $x = 40$: $\hat{y} = 1.5 + 68 = 69.5$ — *not* defensible: it is five times beyond the largest observed spend, where linearity is untested and diminishing returns are likely. Never extrapolate a fitted model far outside the data it was estimated from; with $n = 4$ the fit is fragile in any case.

Outline

Vectors and matrices: the minimum

Objects

A **vector** $\mathrm{a} \in \mathbb{R}^p$ is a column of p numbers; an $m \times n$ **matrix** has m rows and n columns; $\mathrm{A}^\top$ swaps them.

Multiplication — the only rule you must remember

$(m \times n) \cdot (n \times k) = (m \times k)$: the *inner* dimensions must match and they disappear. Entry (i, j) of the product is the dot product of row i of the first and column j of the second.

Special cases

Dot product $\mathrm{a}^\top \mathrm{b} = \sum_j a_j b_j$ (a scalar); identity I with $\mathrm{AI} = \mathrm{A}$; inverse A^{-1} with $\mathrm{A}^{-1}\mathrm{A} = \mathrm{I}$ (exists only if A is square and non-singular).

Dimension checking is free error-detection

If X is $n \times p$, then $\mathrm{X}^\top \mathrm{X}$ is $p \times p$ and $\mathrm{X}^\top \mathrm{y}$ is $p \times 1$. Any formula whose dimensions do not line up is wrong — no algebra needed.

Why this course speaks matrix

Regression in one line

With a column of ones in X, the least-squares estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \qquad \hat{y} = X\hat{\beta}.$$

Read it as a recipe

$X^\top y$ collects how each predictor covaries with the response; $X^\top X$ records how they covary with *each other*; inverting it divides out that shared information — which is what "holding the others constant" means algebraically.

When the inverse fails

If two predictors are perfectly correlated, or $p > n$, then $X^\top X$ is singular and $\hat{\beta}$ does not exist. Ridge regression inverts $X^\top X + \lambda I$ instead (Ch. 6).

Norms, distances and eigenvectors — where they show up

Norms

$$\|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2 \quad (\ell_2), \qquad \|\boldsymbol{\beta}\|_1 = \sum_j |\beta_j| \quad (\ell_1).$$

Where: ridge penalises $\|\boldsymbol{\beta}\|_2^2$ and shrinks coefficients smoothly; lasso penalises $\|\boldsymbol{\beta}\|_1$ and sets some to exactly zero (Ch. 6). The difference between a round and a pointed constraint region is the whole story.

Euclidean distance

$d(\mathrm{a}, \mathrm{b}) = \sqrt{\sum_j (a_j - b_j)^2}$. *Where:* KNN (Ch. 2, 4) and clustering (Ch. 12) are built on it — and it is why unstandardised variables wreck both.

Eigenvectors and eigenvalues

$\mathrm{Av} = \lambda \mathrm{v}$: directions the matrix only stretches. *Where:* principal components (Ch. 6, 12) are the eigenvectors of the covariance matrix — the directions of greatest variance in the data.

Exercise 0.6 — Matrix mechanics [Math]

Exercise

Let

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad y = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}.$$

1. State the dimensions of $X^\top$, $X^\top X$ and $X^\top y$ *before* computing anything.
2. Compute $X^\top X$ and $X^\top y$.
3. Invert $X^\top X$, using $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
4. Compute $\hat{\beta} = (X^\top X)^{-1} X^\top y$, and check the slope against $b_1 = S_{xy}/S_{xx}$.
5. What would go wrong if a third column equal to twice the second were added to X?

Solution — Exercise 0.6

Solution

(1) X is 3×2, so $X^\top$ is 2×3, $X^\top X$ is 2×2 and $X^\top y$ is 2×1; hence $\hat{\beta}$ is 2×1 — an intercept and a slope.

(2)

$$X^\top X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 2+3+5 \\ 2+6+15 \end{pmatrix} = \begin{pmatrix} 10 \\ 23 \end{pmatrix}.$$

(3) $\det = 3(14) - 6(6) = 42 - 36 = 6$, so $(X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$.

(4)

$$\hat{\beta} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}\begin{pmatrix} 10 \\ 23 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 140-138 \\ -60+69 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 2 \\ 9 \end{pmatrix} = \begin{pmatrix} 0.333 \\ 1.5 \end{pmatrix}.$$

Check: $\bar{x} = 2$, $\bar{y} = 10/3$, $S_{xx} = 2$, $S_{xy} = 3$, so $b_1 = 1.5$ ✓ and $b_0 = 10/3 - 3 = 1/3$ ✓.

(5) Perfect collinearity: $X^\top X$ becomes singular ($\det = 0$), the inverse does not exist, and infinitely many coefficient vectors give an identical fit. Software reports a "singular matrix" or silently drops a column.

Outline

Derivatives: the three facts that matter

Rules

$$\frac{d}{dx}x^k = kx^{k-1}, \qquad \frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\log x = \frac{1}{x}, \qquad \frac{d}{dx}f(g(x)) = f'(g(x))\,g'(x).$$

What a derivative means here

The slope of the function: how much the output changes per unit change in the input. At a minimum, the slope is zero — which is how every estimator in this course is derived.

Partial derivatives and the gradient

With several inputs, differentiate with respect to one at a time, holding the others fixed. Stacking them gives the gradient

$$\nabla f(\boldsymbol{\beta}) = \left(\tfrac{\partial f}{\partial \beta_1}, \ldots, \tfrac{\partial f}{\partial \beta_p}\right)^\top,$$

which points in the direction of *steepest increase*.

Minimising a loss: the pattern behind every method

Deriving the sample mean

Minimise $L(c) = \sum_i (x_i - c)^2$ over c:

$$\frac{dL}{dc} = -2\sum_i (x_i - c) = 0 \implies \sum_i x_i = nc \implies c = \bar{x}.$$

So the sample mean is not a convention — it is *the* minimiser of squared error. (Minimising $\sum_i |x_i - c|$ instead gives the median.)

The recurring recipe

1. Write a loss measuring misfit. **2.** Differentiate. **3.** Set to zero and solve — or, when no closed form exists, walk downhill. Least squares, logistic regression, splines, boosting and neural networks are all this recipe with different losses.

Gradient descent: when there is no closed form

The update

$$\beta^{(t+1)} = \beta^{(t)} - \alpha \nabla L\big(\beta^{(t)}\big), \qquad \alpha > 0 \text{ the learning rate.}$$

The picture

Stand on a hillside in fog. The gradient points uphill, so step the opposite way, and repeat. α too small $\Rightarrow$ crawling; α too large $\Rightarrow$ overshooting and divergence. Choosing it well is a recurring practical theme of Chapter 10.

Convexity decides whether this is safe

For a convex loss (least squares, logistic regression) every local minimum is global, so descent finds the answer. Neural networks are *not* convex — which is why initialisation, learning-rate schedules and early stopping matter so much in Chapter 10.

Extended Exercise 0.3 — Normal equations and one descent step [Integrative]

Extended exercise (15 min)

Use the data of Exercise 0.6: $(x, y) = (1, 2), (2, 3), (3, 5)$, model $y = \beta_0 + \beta_1 x$, and loss $L(\beta) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$.

1. Write L out explicitly for the three observations.
2. Derive $\partial L/\partial\beta_0$ and $\partial L/\partial\beta_1$, and show that setting both to zero gives the normal equations $\mathrm{X}^\top \mathrm{X}\boldsymbol{\beta} = \mathrm{X}^\top \mathrm{y}$.
3. Starting from $\boldsymbol{\beta}^{(0)} = (0, 0)^\top$, compute the gradient and take one gradient-descent step with $\alpha = 0.01$.
4. Compute L at $\boldsymbol{\beta}^{(0)}$, at $\boldsymbol{\beta}^{(1)}$, and at the exact solution $(1/3,\ 3/2)^\top$. Did the step help?
5. What happens with $\alpha = 1$? Compute $\boldsymbol{\beta}^{(1)}$ and comment.

Solution — Extended Exercise 0.3 (1/2)

Solution

(1) The loss.

$$L(\beta_0, \beta_1) = (2 - \beta_0 - \beta_1)^2 + (3 - \beta_0 - 2\beta_1)^2 + (5 - \beta_0 - 3\beta_1)^2.$$

(2) Partial derivatives. With $e_i = y_i - \beta_0 - \beta_1 x_i$,

$$\frac{\partial L}{\partial \beta_0} = -2\sum_i e_i, \qquad \frac{\partial L}{\partial \beta_1} = -2\sum_i x_i e_i.$$

Setting both to zero gives $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$, i.e. $3\beta_0 + 6\beta_1 = 10$ and $6\beta_0 + 14\beta_1 = 23$ — exactly $\mathrm{X}^\top \mathrm{X}\boldsymbol{\beta} = \mathrm{X}^\top \mathrm{y}$ with the matrices of Exercise 0.6, so $\boldsymbol{\beta} = (1/3,\ 3/2)^\top$ as before. *The normal equations are "derivative $= 0$" in matrix form.*

(3) One step. At $(0, 0)^\top$ the residuals are $e = (2, 3, 5)$:

$$\nabla L = -2\left(\sum_i e_i,\ \sum_i x_i e_i\right)^\top = -2(10,\ 23)^\top = (-20,\ -46)^\top,$$

$$\boldsymbol{\beta}^{(1)} = (0, 0)^\top - 0.01(-20, -46)^\top = (0.20,\ 0.46)^\top.$$

The step moves *towards* $(0.33, 1.50)$, as it should.

Solution — Extended Exercise 0.3 (2/2)

Solution

(4) Did it help?

$$L(0,0) = 2^2 + 3^2 + 5^2 = 38.$$

At $\beta^{(1)} = (0.20, 0.46)$ the fitted values are $0.66, 1.12, 1.58$, so the residuals are $1.34, 1.88, 3.42$ and

$$L(\beta^{(1)}) = 1.796 + 3.534 + 11.696 = 17.03.$$

At the exact solution the residuals are $1/6, -1/3, 1/6$, so $L = 1/6 \approx 0.167$. One step cut the loss from 38 to 17 — a big improvement, still far from the optimum: descent needs many iterations, which is why a closed form is preferred whenever one exists.

(5) Too large a learning rate. With $\alpha = 1$,

$$\beta^{(1)} = (0,0)^\top - 1 \cdot (-20, -46)^\top = (20,\ 46)^\top.$$

Fitted values are $66, 112, 158$ against $y = (2, 3, 5)$: the loss explodes to about 4.1×10^4, far worse than the starting 38, and the next step is larger still — **divergence**. The step size must respect the curvature of the loss, so in practice one standardises the predictors and tunes α: the workflow of Chapter 10.

Outline

numpy: arrays and vectorised arithmetic

```python
import numpy as np

x = np.array([2, 4, 4, 5, 7, 8, 12]) # a 1-D array
x.mean(), np.median(x), x.std(ddof=1) # 6.0, 5.0, 3.317

z = (x - x.mean()) / x.std(ddof=1) # vectorised: no loop needed
A = np.array([[1, 1], [1, 2], [1, 3]]) # a 3x2 matrix
A.T @ A # matrix product: 2x2
```

The one trap

`np.std()` divides by n by default, not $n-1$. Pass `ddof=1` for the *sample* standard deviation. (pandas uses $n-1$ by default — the two libraries disagree, so state what you mean.)

pandas: the data frame is your data set

```
import pandas as pd

df = pd.read_csv("Wage.csv", index_col=0)
df.shape # (3000, 11) -- that's (n, p)
df.head() # first rows
df.describe() # count, mean, std, min, quartiles, max
df["wage"].median()
df["education"].value_counts() # categorical: counts
df.groupby("education")["wage"].mean() # a summary per group
df[["age", "wage"]].corr() # correlation matrix
df[df["wage"] > 200].shape[0] # filtering rows
```

The four verbs of the labs

select columns, **filter** rows, **group** and **summarise**. Almost every data-preparation step in the thirteen labs is a combination of these.

matplotlib: four plots cover most needs

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(6, 4))
ax.hist(df["wage"], bins=40) # one quantitative variable
ax.boxplot(df["wage"]) # centre, spread, outliers
ax.scatter(df["age"], df["wage"], s=8) # two quantitative variables
ax.bar(counts.index, counts.values) # one categorical variable
ax.set_xlabel("age"); ax.set_ylabel("wage")
plt.show()
```

Plot first — model second

Anscombe's quartet: four data sets with identical means, variances, correlations and regression lines — and completely different shapes. No summary statistic replaces looking at the data.

The two modelling interfaces you will use

```
# statsmodels: inference -- coefficients, SEs, t, p, CIs
import statsmodels.formula.api as smf
fit = smf.ols("sales ~TV", data=adv).fit()
print(fit.summary()) # the table Chapter 3 teaches you to read

# scikit-learn: prediction -- one API for every model
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X_train, y_train)
model.predict(X_test) # .fit / .predict everywhere
```

Which to reach for

`statsmodels` when you want to *understand* (inference: standard errors and *p*-values); `scikit-learn` when you want to *predict* (a uniform `fit/predict` interface and cross-validation tooling). Chapters 3–4 lean on the first, Chapters 5–10 on the second.

Exercise 0.7 — Reading Python output [Python]

Exercise

A student runs the following on the Wage data and gets the output shown:

```
>>> df.shape
(3000, 11)
>>> df["wage"].describe()
# count 3000 | mean 111.70 | std 41.73 | min 20.09
# 25% 85.38 | 50% 104.92 | 75% 128.68 | max 318.34
>>> df["age"].corr(df["wage"])
0.1957
```

1. What are n and p?
2. Compute the IQR. Would a wage of \$220 be flagged as an outlier by the $1.5 \times \mathrm{IQR}$ rule?
3. Compare mean and median. What does the comparison say about the shape, and which should be reported?
4. Interpret the correlation of 0.196. Does it mean age barely matters for wage?
5. Compute the standard error of the mean wage.

Solution — Exercise 0.7

Solution

(1) $n = 3000$ observations, $p = 11$ columns. (If one column is the response `wage`, then 10 are available as predictors.)

(2) $\mathrm{IQR} = 128.68 - 85.38 = 43.30$. The upper fence is $128.68 + 1.5(43.30) = 128.68 + 64.95 = 193.63$. Since $220 > 193.63$, **yes**, it would be flagged — but with $n = 3000$, a few hundred such points are entirely expected in a right-skewed variable. "Flagged" means "look at it", not "drop it".

(3) Mean $111.70 >$ median 104.92, so the distribution is **right-skewed** (a long upper tail of high earners, consistent with max $= 318$ against $Q_3 = 129$). Report the median and IQR alongside the mean — or report the mean and be explicit about the skew.

(4) $r = 0.196$ is a weak *linear* association. It does **not** mean age is unimportant: the wage–age relationship is famously *curved* (rising, then flattening and falling near retirement), and r cannot see curvature. Chapter 7 fits exactly this relationship with polynomials and splines, and finds age to matter a great deal.

(5) $\mathrm{SE}(\bar{x}) = s/\sqrt{n} = 41.73/\sqrt{3000} = 41.73/54.77 \approx 0.76$. So the mean wage is pinned down to well under a dollar, although individual wages vary by ± 40 — the standard-deviation / standard-error distinction in one line.

Exercise 0.8 — Choosing the right tool [Concept]

Exercise

For each task, name (a) the summary or method you would use, and (b) the plot you would draw first.

1. Describe the distribution of house prices in a city.
2. Compare salaries across three departments.
3. Assess whether advertising spend and sales move together.
4. Report how precisely a mean customer rating has been estimated.
5. Judge whether a fitted straight line is adequate.
6. Summarise which of two promotions produced more purchases.

Solution — Exercise 0.8

Solution

1. **Median and IQR** (prices are strongly right-skewed; the mean would be pulled up by a few mansions). Plot: histogram, with a boxplot if outliers are the point.
2. **Group means or medians with SDs**, per department. Plot: side-by-side boxplots — they show centre, spread and outliers in one picture. (A formal comparison would use ANOVA or a regression on dummies, Ch. 3.)
3. **Correlation** r, plus the regression slope if the units matter. Plot: scatterplot *first* — r is meaningless if the pattern is curved.
4. **Standard error** $s/\sqrt{n}$, reported as a confidence interval. Plot: the estimate with an error bar. Note this is the SE, not the SD — precision of the estimate, not spread of the ratings.
5. **Residual plot**: residuals against fitted values. Adequate means a structureless band around zero; curvature means a missing non-linear term, a funnel means non-constant variance (Ch. 3).
6. **Contingency table with row proportions** (purchase rate per promotion), plus a χ^2 test or a difference in proportions. Plot: grouped bar chart of the *rates*, not the raw counts — raw counts mislead whenever the groups differ in size.

Outline

The refresher in one slide

- **Data** are n rows $\times$ p columns; the *type* of the response decides regression vs. classification.
- **One variable**: centre (mean / median), spread (SD / IQR), shape (skew, outliers). Never a centre without a spread.
- **Two variables**: covariance $\to$ correlation r (linear, unit-free, non-causal). Always plot before trusting r.
- **Probability**: conditioning and Bayes' theorem; base rates dominate rare-event problems.
- **Distributions**: Bernoulli/binomial/Poisson for counts; normal, and t, χ^2, F derived from it.
- **Inference**: SD describes data, SE describes an estimate and shrinks like $1/\sqrt{n}$; CIs and p-values are statements about *procedures*.
- **Regression**: minimise RSS; $b_1 = S_{xy}/S_{xx}$; $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$; residual plots diagnose.
- **Algebra and calculus**: dimensions must match; $\hat{\beta} = (\mathrm{X}^\top \mathrm{X})^{-1}\mathrm{X}^\top \mathrm{y}$; set the derivative to zero, or walk downhill.

Key formulas at a glance

Description

$$\bar{x} = \tfrac{1}{n}\sum_i x_i, \quad s^2 = \tfrac{1}{n-1}\sum_i (x_i - \bar{x})^2, \quad r = \tfrac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \quad z_i = \tfrac{x_i - \bar{x}}{s}.$$

Probability and inference

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \quad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X), \quad \mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}},$$

$$\mathsf{CI} = \bar{x} \pm t_{n-1,1-\alpha/2}\,\mathrm{SE}, \qquad t = \frac{\hat{\theta} - \theta_0}{\mathrm{SE}(\hat{\theta})}.$$

Regression and optimisation

$$b_1 = \frac{S_{xy}}{S_{xx}}, \quad b_0 = \bar{y} - b_1\bar{x}, \quad R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \quad \hat{\boldsymbol{\beta}} = (\mathsf{X}^\top\mathsf{X})^{-1}\mathsf{X}^\top\mathsf{y},$$

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \alpha\nabla L(\boldsymbol{\beta}^{(t)}).$$

Vocabulary check

Term	One-line meaning
Population / sample	all units of interest / the rows you have
Parameter / statistic	unknown truth (μ) / computed from data ($\bar{x}$)
Standard deviation	spread of the *data*
Standard error	spread of an *estimate* across samples
Quartile, IQR	25%/75% points; their difference
Skewness	asymmetry; mean pulled towards the long tail
z-score	value expressed in standard deviations from the mean
Covariance / correlation	joint movement; its unit-free version
Confounder	third variable driving both X and Y
Conditional probability	probability given that something is known
Base rate	the unconditional prevalence $P(A)$
Type I / Type II error	false positive / false negative
Power	probability of detecting a real effect
Residual	observed minus fitted, $y_i - \hat{y}_i$
R^2	fraction of variation in y accounted for
Gradient	vector of partial derivatives; points uphill
Convex	bowl-shaped; every local minimum is global

Where each topic reappears in the course

Refresher topic	**Comes back in**
Variable types, tidy data	Ch. 1–2 (notation), Ch. 3 (dummy variables)
Mean, variance, expectation rules	Ch. 2 (bias–variance decomposition)
Standardisation (z-scores)	Ch. 2 (KNN), Ch. 6 (ridge/lasso), Ch. 10
Correlation, confounding	Ch. 3 (multiple regression, collinearity)
Conditional probability, Bayes	Ch. 4 (Bayes classifier, LDA, naive Bayes)
Bernoulli / binomial / Poisson	Ch. 4 (logistic, Poisson regression)
Normal, t, χ^2, F	Ch. 3 (t- and F-tests on coefficients)
Standard errors, CIs	Ch. 3 (coefficient inference), Ch. 5 (bootstrap)
Type I/II errors, p-values	Ch. 4 (confusion matrix), Ch. 13 (FWER, FDR)
Simple linear regression, R^2	Ch. 3 (all of it), Ch. 6 (model selection)
Residual plots	Ch. 3 (diagnostics), Ch. 7 (non-linearity)
Matrix algebra, inverses	Ch. 3 (normal equations), Ch. 6 (singularity)
ℓ_1 / ℓ_2 norms	Ch. 6 (lasso vs. ridge geometry)
Eigenvectors	Ch. 6 (PCR), Ch. 12 (PCA)
Derivatives, gradients, convexity	Ch. 10 (backpropagation, training)
`pandas` / `sklearn`	every lab notebook

Common pitfalls to leave behind

Don't

1. Report a mean for a skewed variable without the median — or any centre without a spread.
2. Trust r without looking at the scatterplot.
3. Read a correlation, or a regression coefficient, as a causal effect.
4. Confuse the standard deviation with the standard error.
5. Say "$p > 0.05$, therefore no effect", or "p small, therefore important".
6. Delete outliers because a rule flagged them.
7. Extrapolate a fitted model beyond the range of the data.
8. Compare distances or penalise coefficients on unstandardised variables.

Self-check questions

1. Why does the sample variance divide by $n - 1$?
2. Give two variables with a strong relationship but $r \approx 0$.
3. A test is 99% accurate for a disease affecting 1 in 10,000. You test positive. Estimate $P(\text{ill} \mid +)$ — roughly.
4. State precisely what “95% confident” means.
5. Why does quadrupling the sample size only halve the standard error?
6. In $\hat{y} = 4 + 0.8x$ with x in years and y in €000s, interpret both coefficients with units.
7. If X is 200×5, what size is $(\mathrm{X}^\top \mathrm{X})^{-1}\mathrm{X}^\top \mathrm{y}$?
8. Why must predictors be standardised before ridge regression?
9. Why does a large learning rate make gradient descent diverge?

Five things to remember

1. **Plot before you compute.** Every summary statistic hides something; a picture usually shows it.
2. **Centre needs spread; estimate needs uncertainty.** A number without a standard error is not a result.
3. **Conditioning is the core idea.** Classification is estimating $P(Y \mid X)$, and base rates matter enormously.
4. **Association is not causation.** Ask what third variable could produce the pattern — that question is why multiple regression exists.
5. **Fitting is minimising a loss.** Differentiate and solve, or walk downhill. The rest of the course changes the loss and the model, not the recipe.

Where to read more

If you want to go deeper before Lecture 1

- **ISLP Chapter 2** — assessing model accuracy; the bias–variance trade-off. It picks up exactly where this deck stops.
- **ISLP Chapter 3.1** — simple linear regression, with the inference (standard errors, t-tests) that we only sketched here.
- Any introductory statistics text, for: descriptive statistics, probability rules, the normal and t distributions, confidence intervals, hypothesis tests, and simple regression.
- A first linear-algebra text, for: matrix multiplication, inverses, and eigenvectors.

Practice beats reading

Redo Exercises 0.2, 0.3 and Extended Exercise 0.2 *by hand*, then reproduce them in Python with `numpy` and `pandas`. That single hour is the best preparation for Lecture 1.

What's next

Lecture 1 — *Chapter 1: Introduction* and the start of *Chapter 2: Statistical Learning*.

- What statistical learning is, and why $Y = f(X) + \varepsilon$.
- Prediction versus inference — two different goals, two different modelling strategies.
- Supervised versus unsupervised, regression versus classification.
- The course data sets: `Wage`, `Smarket`, `NCI60`.

Bring with you

The vocabulary of this deck. From Lecture 1 on, "standard error", "residual", "conditional probability" and "gradient" are used without further explanation.